

## CHANGJON, Korea, Democratic People's Republic of

WMO: 470610

Lat: 38.733N

Lon: 128.183E

Elev: 112

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 16.6       | 19.9 | -6.1                          | 4.0 | 22.8 | -1.6 | 5.1 | 24.2 | 25.7                  | 30.2 | 20.9 | 30.1 | 7.7                   | 320  | 0.369 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.9                    | 89.0            | 78.1 | 86.8 | 76.8 | 84.4 | 75.5 | 81.0                | 86.2 | 79.3 | 84.1 | 77.6 | 82.1 | 5.3                  | 90   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 79.6                            | 154.1 | 84.6 | 77.9 | 145.4 | 82.6 | 76.3 | 137.9 | 80.8 | 44.9          | 86.3 | 43.0 | 84.3 | 41.3 | 82.6 | 89.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |    |    |
| 23.8              | 19.3 | 15.6 | 11.4                       | 93.0 | 5.1                | 3.2 | 7.8                                                | 95.3 | 4.8        | 97.2 | 2.0        | 99.0 | -1.7       | 101.3 |    |    |
|                   |      |      | 8.7                        | 83.4 | 5.0                | 2.6 | 5.1                                                | 85.2 | 2.2        | 86.7 | -0.5       | 88.2 | -4.1       | 90.0  |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 54.5   | 31.3  | 33.3  | 41.9  | 53.2  | 62.6  | 68.5  | 75.2  | 75.9  | 68.3  | 59.5  | 47.4  | 36.1 |
|                                                                                  | DBStd       | 16.72  | 6.11  | 5.90  | 7.32  | 7.31  | 6.91  | 5.40  | 6.01  | 4.79  | 4.12  | 5.00  | 7.15  | 6.94 |
|                                                                                  | HDD50       | 1937   | 581   | 467   | 273   | 47    | 1     | 0     | 0     | 0     | 0     | 4     | 132   | 432  |
|                                                                                  | HDD65       | 4782   | 1046  | 886   | 717   | 359   | 129   | 23    | 2     | 0     | 15    | 179   | 529   | 897  |
|                                                                                  | CDD50       | 3597   | 0     | 0     | 21    | 144   | 393   | 555   | 781   | 802   | 549   | 297   | 54    | 1    |
|                                                                                  | CDD65       | 965    | 0     | 0     | 0     | 6     | 56    | 128   | 318   | 337   | 113   | 7     | 0     | 0    |
|                                                                                  | CDH74       | 6557   | 0     | 0     | 0     | 33    | 376   | 732   | 2624  | 2442  | 341   | 9     | 0     | 0    |
|                                                                                  | CDH80       | 1848   | 0     | 0     | 0     | 4     | 69    | 166   | 900   | 672   | 37    | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.8    | 4.9   | 4.8   | 5.9   | 6.6   | 5.8   | 3.6   | 3.7   | 3.6   | 4.0   | 4.4   | 5.2   | 5.4  |
| Precipitation                                                                    | PrecAvg     | 63.0   | 1.7   | 2.1   | 1.8   | 3.0   | 4.1   | 6.1   | 12.9  | 12.5  | 8.4   | 4.1   | 4.1   | 2.0  |
|                                                                                  | PrecMax     | 92.9   | 5.0   | 8.1   | 4.0   | 6.1   | 13.0  | 18.1  | 23.9  | 26.2  | 21.3  | 13.5  | 28.6  | 10.1 |
|                                                                                  | PrecMin     | 43.7   | 0.1   | 0.0   | 0.2   | 0.4   | 0.7   | 1.5   | 3.1   | 4.0   | 1.0   | 0.2   | 0.4   | 0.0  |
|                                                                                  | PrecStd     | 11.6   | 1.4   | 2.0   | 1.3   | 1.7   | 2.8   | 4.6   | 5.6   | 5.9   | 5.3   | 3.5   | 5.4   | 2.5  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 49.6  | 53.9  | 68.4  | 78.5  | 85.3  | 88.3  | 91.5  | 91.8  | 83.7  | 75.4  | 68.3  | 56.1 |
|                                                                                  |             | MCWB   | 41.3  | 43.3  | 54.1  | 60.7  | 63.9  | 71.2  | 78.0  | 81.2  | 73.3  | 63.6  | 57.0  | 48.3 |
|                                                                                  | 2%          | DB     | 45.4  | 49.0  | 61.8  | 73.4  | 81.5  | 84.4  | 89.3  | 88.9  | 80.5  | 72.0  | 64.1  | 52.3 |
|                                                                                  |             | MCWB   | 38.2  | 40.9  | 49.1  | 57.0  | 62.5  | 70.1  | 78.1  | 79.9  | 71.2  | 62.8  | 54.7  | 45.7 |
|                                                                                  | 5%          | DB     | 42.8  | 45.6  | 58.1  | 69.6  | 78.2  | 81.2  | 87.4  | 86.3  | 77.6  | 69.8  | 61.2  | 48.6 |
|                                                                                  |             | MCWB   | 36.6  | 38.3  | 47.5  | 55.0  | 61.0  | 69.2  | 77.5  | 77.9  | 69.9  | 62.2  | 53.3  | 42.8 |
|                                                                                  | 10%         | DB     | 40.0  | 42.4  | 54.0  | 66.1  | 74.7  | 78.4  | 85.3  | 83.7  | 75.4  | 67.8  | 58.1  | 45.8 |
|                                                                                  |             | MCWB   | 34.1  | 36.2  | 44.9  | 53.4  | 59.8  | 68.5  | 76.8  | 77.3  | 69.3  | 60.9  | 51.7  | 40.3 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 42.2  | 44.7  | 56.0  | 64.4  | 69.5  | 76.4  | 82.5  | 84.2  | 76.5  | 68.2  | 60.0  | 50.4 |
|                                                                                  |             | MCDB   | 46.7  | 51.7  | 65.2  | 73.5  | 80.3  | 83.2  | 87.6  | 88.2  | 81.2  | 71.2  | 63.9  | 54.2 |
|                                                                                  | 2%          | WB     | 39.5  | 41.6  | 51.2  | 60.1  | 66.4  | 73.8  | 80.6  | 81.8  | 73.8  | 65.9  | 56.7  | 46.5 |
|                                                                                  |             | MCDB   | 44.3  | 47.6  | 59.6  | 69.1  | 76.2  | 79.1  | 86.6  | 86.8  | 77.6  | 69.4  | 61.2  | 51.3 |
|                                                                                  | 5%          | WB     | 37.0  | 39.3  | 48.0  | 57.2  | 64.2  | 72.0  | 79.1  | 80.1  | 72.1  | 64.1  | 54.7  | 43.7 |
|                                                                                  |             | MCDB   | 41.6  | 44.4  | 56.1  | 66.2  | 73.0  | 77.6  | 85.0  | 84.5  | 75.6  | 68.0  | 59.4  | 48.0 |
|                                                                                  | 10%         | WB     | 35.1  | 37.2  | 45.5  | 54.6  | 62.3  | 70.3  | 77.6  | 78.5  | 70.6  | 62.4  | 52.5  | 41.0 |
|                                                                                  |             | MCDB   | 38.8  | 41.4  | 52.9  | 63.4  | 70.3  | 75.7  | 83.2  | 82.1  | 73.9  | 66.2  | 57.3  | 45.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.4  | 11.0  | 11.7  | 13.7  | 13.8  | 10.9  | 8.9   | 8.9   | 11.0  | 12.3  | 11.7  | 11.5 |
|                                                                                  |             | MCDBR  | 13.4  | 14.8  | 17.4  | 20.2  | 19.9  | 17.5  | 12.8  | 12.4  | 14.0  | 14.9  | 15.0  | 13.5 |
|                                                                                  |             | MCWBR  | 10.2  | 10.6  | 11.1  | 11.2  | 10.1  | 9.4   | 7.3   | 7.4   | 9.2   | 10.1  | 10.4  | 10.8 |
|                                                                                  | 5% WB       | MCDBR  | 12.0  | 13.7  | 16.1  | 18.3  | 17.6  | 13.8  | 11.7  | 10.8  | 11.3  | 12.9  | 12.6  | 12.8 |
|                                                                                  |             | MCWBR  | 9.7   | 10.5  | 11.4  | 11.9  | 10.8  | 9.0   | 7.8   | 7.9   | 8.6   | 9.8   | 9.8   | 10.8 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.394  | 0.453 | 0.532 | 0.592 | 0.599 | 0.648 | 0.615 | 0.537 | 0.475 | 0.476 | 0.440 | 0.390 |      |
|                                                                                  | taud        | 2.032  | 1.903 | 1.784 | 1.707 | 1.744 | 1.706 | 1.843 | 2.047 | 2.144 | 2.004 | 2.024 | 2.076 |      |
|                                                                                  | Ebn at Noon | 233    | 234   | 229   | 225   | 227   | 216   | 222   | 237   | 244   | 225   | 215   | 224   |      |
|                                                                                  | Edh at Noon | 39     | 51    | 64    | 73    | 72    | 75    | 65    | 52    | 44    | 46    | 40    | 35    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 771    | 1030  | 1315  | 1571  | 1725  | 1636  | 1353  | 1353  | 1266  | 1078  | 751   | 683   |      |
|                                                                                  | RadStd      | 69     | 95    | 122   | 134   | 179   | 143   | 185   | 157   | 145   | 94    | 103   | 68    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (29 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +187  | +99   |

Nomenclature: See separate page